

1.5 SET CONSTRUCTION

Example 1.5.1

Let S = Set of all people
 B = Set of all business men
 C = Set of all people who smoke cigarettes
 D = Set of all pipe smokers
 E = Set of all wine drinkers

How do we express the statement "Some businessmen pipe smokers are wine drinkers" in set notation? The answer is $B \cap D \cap E \neq \phi$. The statement "A businessman is neither a wine drinker nor a pipe smoker" is expressed in set notation as $B \subset (D' \cap E')$.

The following example is a typical one illustrating the use of set theory.

Example 1.5.2

In a survey of 100 students in a music school, the number of students learning different musical instruments were found to be: guitar 28; veena 30; flute 42; guitar and veena 8; guitar and flute 10; veena and flute 5; all musical instruments 3.

- (a) How many students were learning none of these three musical instruments?
(b) How many students were learning only the flute?

The Venn diagram is shown in Fig. 1.5.1.

Let G , V and F represent the set of students learning guitar, veena and flute respectively. The set of students learning none of the three instruments is $G' \cap F' \cap V'$ and the set of students learning flute only is

$F \cap V' \cap G'$. We would like to know how many elements are there in these two sets.

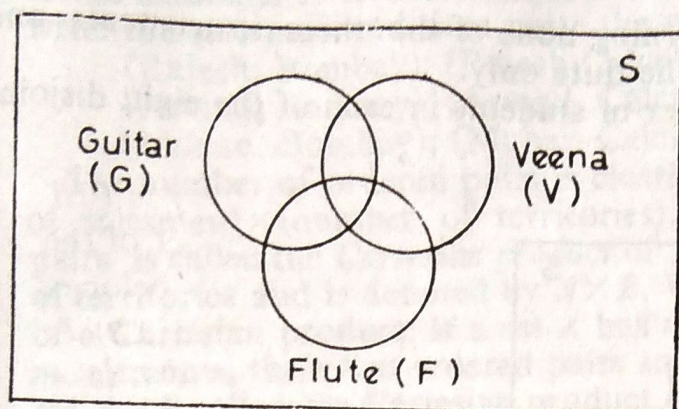


Fig. 1.5.1

For the given three sets G , V and F we can, in the Venn diagram, distinguish eight disjoint sets I-VIII (Fig. 1.5.2).

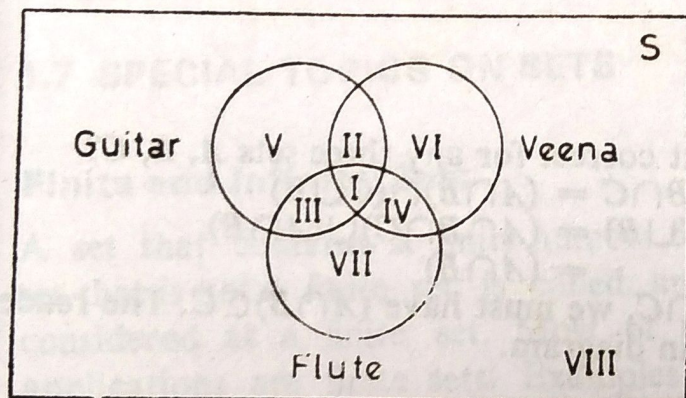


Fig. 1.5.2

Using the data given in the example we have to find the number of elements in each of these disjoint sets. We proceed systematically as follows.

Set I is $G \cap F \cap V$ and is the set of all students learning all the three musical instruments and it is given that the number of elements in this set is 3. Set II is the set $G \cap V \cap F'$ and is the set of students learning the guitar and veena but not the flute. The number of students belonging to this set is the difference of "the number of students who are learning the guitar and veena" and "the number of students learning all the three instruments" and is therefore equal to $8 - 3 = 5$. By a similar reasoning,

The number of students in III = $10 - 3 = 7$

The number of students in IV = $5 - 3 = 2$

Since the total number of students learning the guitar is 28, the number of students in set V: $G \cap V' \cap F'$ is $28 - 3 - 5 - 7 = 13$. Similarly, the number of students in VI: $V \cap G' \cap F'$ is $30 - 3 - 5 - 2 = 20$ and the number of students in set VII: $F \cap V' \cap G'$ is $42 - 3 - 7 - 2 = 30$. Finally the number of elements in set VIII: $G' \cap F' \cap V'$ is obtained as

Total number of students—the number of students in I to VII
 $= 100 - 3 - 5 - 7 - 2 - 13 - 20 - 30$
 $= 20$

Thus, there are 20 students learning none of the three instruments and 30 students (set VII) learning the flute only.
 Figure 1.5.3 shows the number of students in each of the eight disjoint sets.

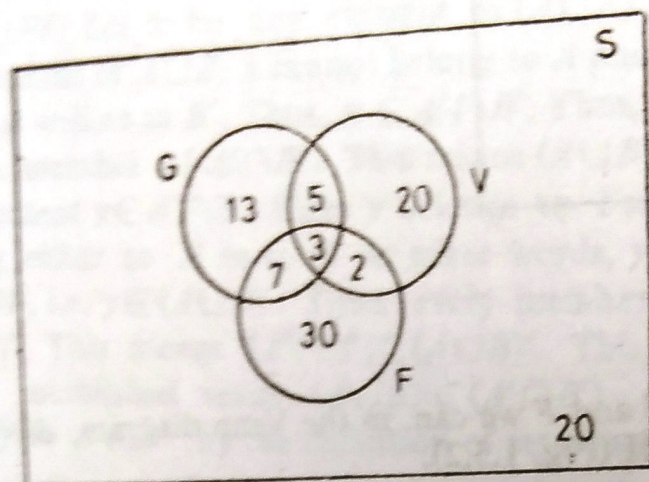


Fig. 1.5.3

Example 1.5.3

When is the following statement correct for any three sets A, B, C ?

$$A \cap B \cap C = (A \cap B) \cap (C \cup B)$$

Solution:

$$\begin{aligned} A \cap B \cap (C \cup B) &= (A \cap B \cap C) \cup (A \cap B) \\ &= (A \cap B) \end{aligned}$$

If this were equal to $A \cap B \cap C$, we must have $(A \cap B) \subset C$. The reader can verify this through the Venn diagram.

1.6 CARTESIAN PRODUCT

Before we define the operation of obtaining a new set called the 'Cartesian product' of two given sets, let us introduce the notion of an 'ordered pair' of elements. Suppose we have two numbers a and b . The pair of numbers can be represented as a set (a, b) or (b, a) if the order of writing is not important. If the order is important, i.e. if we want to distinguish (a, b) from (b, a) , we can refer to an ordered pair or ordered set. As an example, suppose, in a family with a boy and a girl, we specify the sex of the children by a set. We can write (Boy, Girl) or (Girl, Boy). If, however, we want to write the sex of the children according to the order of their birth, we write the set as (Boy, Girl) if the boy is older, or as (Girl, Boy) if the girl is older.

In the latter case, (Boy, Girl) becomes an ordered pair, the order being specified by the order of birth.

Example 1.6.1

Suppose that set A is the set of salesmen (Rajesh, Arvind, Mohan) and another set B denotes the territories (Bombay, Calcutta), we can write

all the possible ordered pairs which will indicate the possible allocations of the salesmen to territories, such that in each ordered pair we first write the salesman and then write the name of the territory:

(Rajesh, Bombay); (Rajesh Calcutta);
 (Arvind, Bombay); (Arvind, Calcutta);
 (Mohan, Bombay); (Mohan Calcutta).

The number of ordered pairs is clearly 6 being equal to the (number of salesmen) $\times$ (number of territories). The set of all these six ordered pairs is called the *Cartesian product* of $A =$ set of salesmen and $B =$ set of territories and is denoted by $A \times B$. We can state the general definition of a Cartesian product: If a set A has n elements and another set B has m elements, the $n \times m$ ordered pairs such that $x \in A$ and $y \in B$ form a set $A \times B$ called the Cartesian product of A and B is given by

$$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$$

Note that $A \times B$ is not necessarily the same as $B \times A$, since we are dealing with ordered pairs. We can also talk of a Cartesian product $A \times A$, if A represents the set of the outcomes when a coin is tossed so that $A = \{\text{Heads, Tails}\}$, $A \times A = \{(\text{Heads, Heads}); (\text{Heads, Tails}); (\text{Tails, Heads}), (\text{Tails, Tails})\}$.

1.7 SPECIAL TOPICS ON SETS

Finite and Infinite Sets

A set that contains a finite number of elements is called a *finite set*. A set that is not a finite set is called an *infinite set*. The empty set ϕ is considered as a finite set. Most of the sets that one encounters in applications are finite sets. Examples of an infinite set are the set of all positive integers, the set of all points in an interval, etc.

Number of Elements in a Set

Let us denote the number of elements of a set A by $N(A)$. Suppose, A is the set of all skilled workers and B is the set of all workers who are on the permanent roll, we see from Venn diagram (Fig. 1.7.1)

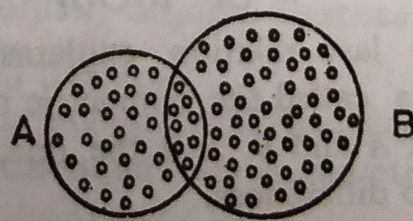


Fig. 1.7.1

that

$$N(A \cup B) = N(A) + N(B) - N(A \cap B).$$

In the above formula, we subtract $N(A \cap B)$ from $N(A) + N(B)$ because the elements in $A \cap B$ get counted twice, once in A and again in B . A similar formula can be given for $N(A \cup B \cup C)$ for any sets A , B and C :

$$N(A \cup B \cup C) = N(A) + N(B) + N(C) - N(A \cap B) - N(A \cap C) - N(B \cap C) + N(A \cap B \cap C)$$

The reader is asked to verify the above formula through Venn diagram and suggest a similar formula for $N(A \cup B \cup C \cup D)$ when there are four sets A, B, C and D .

Example 1.7.1

A survey was conducted in a region to study the radio listening habits and cinema-viewing habits of the inhabitants. The following data were obtained. The figures are in thousands.

Number of people who listen to Radio Programme I regularly : 37,055

Number of people who listen to Radio Programme II regularly : 29,272

Number of people who see Cinema regularly : 52,606

Number of people who listen to both the radio programmes regularly : 15,078

Radio Programme I and Cinema : 27,341

Radio Programme II and Cinema : 26,272

Radio Programmes I and II and Cinema : 9,787

How many people do at least one thing?

Let A = Set of people who listen to Radio Programme I regularly

B = Set of people who listen to Radio Programme II regularly

C = Set of people who view Cinema regularly.

Applying the formula for $N(A \cup B \cup C)$,

$$\begin{aligned} N(A \cup B \cup C) &= 37,055 + 29,272 + 52,606 - 15,078 \\ &\quad - 27,341 - 26,272 + 9,787 \\ &= 60,029 \end{aligned}$$

Thus, 60,029 do at least one of the three things. The reader can verify the answer by drawing the Venn diagram and calculating $N(A \cup B \cup C)$.

Symmetric Difference of Two Sets

Let A and B be any two sets. The symmetric difference of two sets A and B is the set of elements that belong to only A or only to B and is denoted by $A \Delta B$. In symbols

$$\begin{aligned} A \Delta B &= (x \mid x \in A \cap B' \text{ or } x \in A' \cap B) \\ &= (A \cap B') \cup (A' \cap B) \\ &= (A - B) \cup (B - A) \end{aligned}$$

This last relation explains why $A \Delta B$ is called the symmetric difference of A and B . It is the union of the two differences.

$A \Delta B$ is shown by the shaded area in Fig. 1.7.2.

It can easily be verified that

$$A \cup B = (A \Delta B) \cup (A \cap B)$$

and that $A \Delta B$ and $A \cap B$ are disjoint.

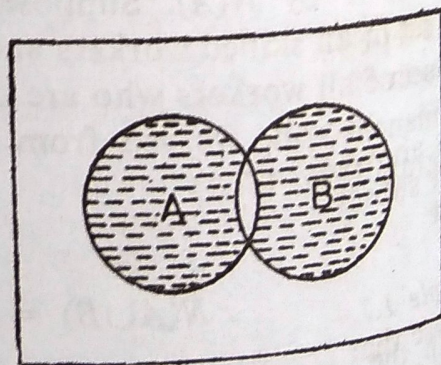


Fig. 1.7.2

Example 1.7.2

A manufactured item can have a defect in its shape or in its weight. A sample of 100 items showed that 10 of them had defect in its shape and 12 of them in weight. Five of them had both the defects. How many of them had exactly one defect?

Let set A = Set of items with defect in shape, and
 set B = Set of items with defect in weight.

We are given $N(A) = 10$;
 $N(B) = 12$; $N(A \cap B) = 5$.
 We are required to find out $N(A \Delta B)$. The Venn diagram is shown in Fig. 1.7.3.

$$\begin{aligned} N(A \Delta B) &= N(A - B) + N(B - A) \\ &= 5 + 7 \\ &= 12 \end{aligned}$$

Thus 12 items had exactly one defect.

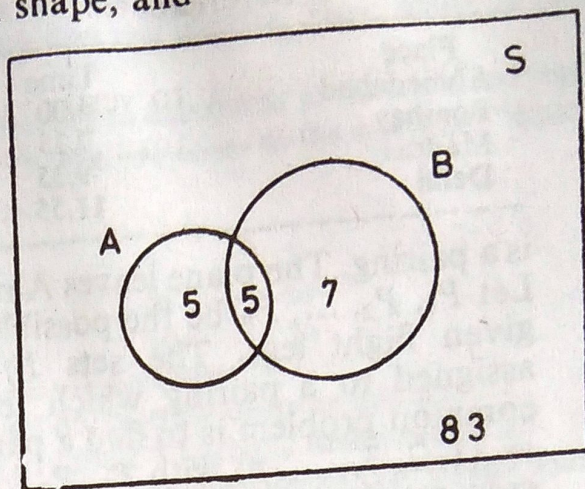


Fig. 1.7.3

Partition of Set

Consider any set H . A *partition* of H is defined as the sets $H_1, H_2, \dots, H_n$ such that these sets are mutually disjoint, i.e. $H_i \cap H_j = \phi$ for any i and j ($i \neq j$) and $H_1 \cup H_2 \cup \dots \cup H_n = H$.

See Fig. 1.7.4 for the case $n = 5$.

Here $H = H_1 \cup H_2 \cup H_3 \cup H_4 \cup H_5$.
 H_i 's are mutually disjoint. The set H is partitioned into five sets H_1, H_2, H_3, H_4 and H_5 .

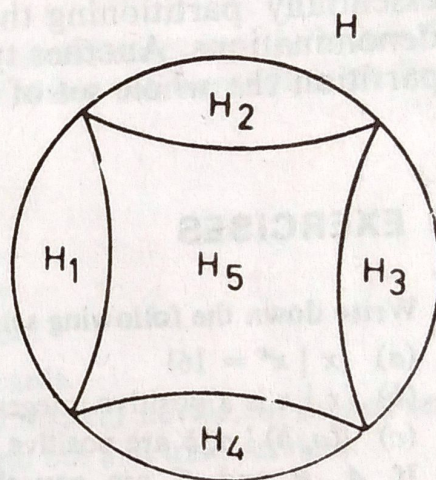


Fig. 1.7.4

Example 1.7.3

Suppose H is the set of all people in Ahmedabad. A partition of H can be effected by classifying them according to their religions, say for example, Hindus (H_1), Muslims (H_2), Christians (H_3), and others (H_4). Then $H = H_1 \cup H_2 \cup H_3 \cup H_4$ and these are mutually disjoint. Suppose, F is the set of all females in Ahmedabad. We can write

$$F = (H_1 \cap F) \cup (H_2 \cap F) \cup (H_3 \cap F) \cup (H_4 \cap F)$$

This set-equality expression merely states that the set of all females is the union of Hindu females, Muslim females, Christian females and other females.

The above example shows that if $H_1, H_2, \dots, H_n$ is a partition of a set H , any subset B of H can be expressed as

$$B = (H_1 \cap B) \cup (H_2 \cap B) \dots \cup (H_n \cap B)$$

Example 1.7.4

Suppose that an airline has to operate a number of flight legs 1, 2, ..., n during a week. A flight leg is the flight of an airplane from one town to another. For example, the flight of a plane leaving Ahmedabad and landing at Bombay is a flight leg. A round trip of flight legs that an airplane can perform is called a *pairing*. For example,